

1. (10 points) Suppose $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation which satisfies

$$T\left(\begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T\left(\begin{bmatrix} 2 \\ 10 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

Find a *nonzero* vector $\mathbf{v} \in \mathbb{R}^3$ such that $T(\mathbf{v}) = \mathbf{0}$.

2. (a) (3 points) Give three distinct 2×2 matrices A with real entries such that $A^3 = I_2$. (*Hint:* Consider rotation matrices. You may express the entries using cosines and sines.)
- (b) (7 points) Give seven distinct 3×3 matrices A with real entries such that $A^3 = I_3$. (*Hint:* Consider block matrices / rotations about the three coordinate axes.)